Ziqiao Wang

Associate Professor

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Employment

Jun. 2026 - Now **Deputy Chair**, *Department of Cybersecurity*, School of Computer Science and Technology, Tongji University, *Shanghai, China*.

Dec. 2025 - Now **Associate Professor**, School of Computer Science and Technology, Tongji University, *Shanghai, China*.

Sep. 2024 - Dec. 2025 **Assistant Professor**, School of Computer Science and Technology, Tongji University, *Shanghai, China*.

May 2019 - Dec. 2019 **Research Intern**, Digital Technologies Research Centre, National Research Council Canada (NRC), *Ottawa, Canada*.
Mentor: Dr. Harry H.Y. Guo

Education

Jan. 2019 - May 2024 **Ph.D, University of Ottawa**, *Ottawa, Ontario, Canada*.
Electrical and Computer Engineering
Thesis: "Generalization in Machine Learning through Information-Theoretic Lens"
Advisor: Yongyi Mao
Committee: Maia Fraser, Tommaso R. Cesari, James R. Green, Ashish Khisti

Sep. 2017 - Dec. 2018 **M.A.Sc, University of Ottawa**, *Ottawa, Ontario, Canada*.
Electrical and Computer Engineering
Advisor: Yongyi Mao

Sep. 2013 - Jun. 2017 **B.Eng, North China Electric Power University (NCEPU)**, *Baoding, Hebei, China*.
Electrical Engineering and its Automation
Advisor: Xiangyu Zhang

Research Interests

- **Area:** Machine Learning, Statistical Learning Theory, Information Theory, LLMs
- **Topics:** Generalization, LLM Reasoning, LLM Quantization, Domain Adaptation

Publications & Preprints

$*$ denotes equal contribution. † denotes corresponding author(s).

Preprints

P1 Fanshuang Kong, Richong Zhang, Zhijie Nie, **Ziqiao Wang** and Qiang Sun. "Fine-tuning Aligned Classifiers for Merging Outputs: Towards a Superior Evaluation Protocol in Model Merging." *Submitted.*

P2 Fanshuang Kong, Richong Zhang and **Ziqiao Wang**. "Activated Parameter Locating via Causal Intervention for Model Merging." *Submitted.*

P3 **Ziqiao Wang**, Yongyi Mao, Hongyu Guo and Richong Zhang. "On SkipGram Word Embedding Models with Negative Sampling: Unified Framework and Impact of Noise Distributions." *arXiv preprint arXiv:2009.04413, 2020.*

Conference Proceedings

C1 Gengze Xu, Wei Yao, **Ziqiao Wang**†, and Yong Liu†. "Weak-to-Strong Generalization via Bregman BiasVariance Decomposition." *International Conference on Machine Learning (ICML), 2026.*

C2 Tianpei Zou, Sanqing Qu, Guo Yu, Ya Wu, Fan Lu, Zhongcong Xu, Zhang Bo, **Ziqiao Wang** and Guang Chen. "GOOD: Geometry-guided Out-of-Distribution Modeling for Open-set Test-time Adaptation in Point Cloud Semantic Segmentation." *International Conference on Learning Representations (ICLR), 2026.*

C3 Qingjun Wang, Hongtu Zhou, Hang Yu, Junqiao Zhao, Yanping Zhao, Chen Ye, **Ziqiao Wang** and Guang Chen. "Beyond Penalization: Diffusion-based Out-of-Distribution Detection and Selective Regularization in Offline Reinforcement Learning." *International Conference on Learning Representations (ICLR), 2026.*

C4 Chang Huang, Shatong Zhu, Junqiao Zhao, Hongtu Zhou, Hai Zhang, Di Zhang, Chen Ye, **Ziqiao Wang** and Guang Chen. "Potentially Optimal Joint Actions Recognition for Cooperative Multi-Agent Reinforcement Learning." *International Conference on Learning Representations (ICLR), 2026.*

C5 Haiyun He, Yepeng Liu, **Ziqiao Wang**, Yongyi Mao and Yuheng Bu. "Theoretically Grounded Framework for LLM Watermarking: A Distribution-Adaptive Approach." *Advances in Neural Information Processing Systems (NeurIPS), 2025.*

C6 Ao Zhou, Jiayi Guan, Li Shen, Fan Lu, Sanqing Qu, Junqiao Zhao, **Ziqiao Wang**, Ya Wu, Guang Chen. "CHPO: Constrained Hybrid-action Policy Optimization for Reinforcement Learning." *Advances in Neural Information Processing Systems (NeurIPS), 2025.*

C7 Qiwen Gu, Xufei Wang, Junqiao Zhao, Siyue Tao, Tiantian Feng, **Ziqiao Wang**, Guang Chen. "MutualVPR: A Mutual Learning Framework for Resolving Supervision Inconsistencies via Adaptive Clustering." *Advances in Neural Information Processing Systems (NeurIPS), 2025.*

C8 Haiyun He, Yepeng Liu, **Ziqiao Wang**, Yongyi Mao, and Yuheng Bu. "Distributional Information Embedding: A Framework for Multi-bit Watermarking." *Proceedings of the Asia Pacific Workshop on Data Science and Information Theory (APWDSIT), 2025.*

C9 Wei Yao, Wenkai Yang, **Ziqiao Wang**, Yankai Lin, and Yong Liu. "Revisiting Weak-to-Strong Generalization in Theory and Practice: Reverse KL vs. Forward KL." *Findings of the Association for Computational Linguistics: ACL 2025.*

C10 Peiyi Zhang, Richong Zhang, Zhijie Nie and **Ziqiao Wang**. "Dynamic Task Vector Grouping for Efficient Multi-Task Prompt Tuning." *Findings of the Association for Computational Linguistics: ACL 2025.*

C11 **Ziqiao Wang**, Cheng Long and Yongyi Mao. "Generalization in Federated Learning: A Conditional Mutual Information Framework." *International Conference on Machine Learning (ICML), 2025.*

C12 Fanshuang Kong, Richong Zhang, Xiaohui Guo, Junfan Chen and **Ziqiao Wang**. "Preserving Label Correlation for Multi-label Text Classification by Prototypical Regularizations." *Proceedings of the ACM on Web Conference (WWW), 2025.*

C13 Fanshuang Kong, Richong Zhang and **Ziqiao Wang**. "LH-Mix: Local Hierarchy Correlation Guided Mixup over Hierarchical Prompt Tuning." *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2025*

C14 **Ziqiao Wang** and Yongyi Mao. "Generalization Bounds via Conditional f-Information." *Advances in Neural Information Processing Systems (NeurIPS), 2024.*

C15 **Ziqiao Wang** and Yongyi Mao. "On f-Divergence Principled Domain Adaptation: An Improved Framework." *Advances in Neural Information Processing Systems (NeurIPS), 2024.*

C16 **Ziqiao Wang** and Yongyi Mao. "Two Facets of SDE Under an Information-Theoretic Lens: Generalization of SGD via Training Trajectories and via Terminal States." *Proceedings of the Fortieth Conference on Uncertainty in Artificial Intelligence (UAI), 2024.*

C17 Fanshuang Kong, Richong Zhang, **Ziqiao Wang** and Yongyi Mao. "On Unsupervised Domain Adaptation: Pseudo Label Guided Mixup for Adversarial Prompt Tuning." *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI), 2024.*

C18 Hailang Huang, Zhijie Nie, **Ziqiao Wang** and Ziyu Shang. "Cross-modal and Uni-modal Soft-label Alignment for Image-Text Retrieval." *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI), 2024.*

C19 **Ziqiao Wang** and Yongyi Mao. "Sample-Conditioned Hypothesis Stability Sharpens Information-Theoretic Generalization Bounds." *Advances in Neural Information Processing Systems (NeurIPS), 2023.*

C20 **Ziqiao Wang** and Yongyi Mao. "Tighter Information-Theoretic Generalization Bounds from Supersamples." *International Conference on Machine Learning (ICML), 2023* **(Oral Presentation, top 2.2% of submissions)**.

C21 Zixuan Liu*, **Ziqiao Wang***, Hongyu Guo and Yongyi Mao. "Over-Training with Mixup May Hurt Generalization." *International Conference on Learning Representations (ICLR), 2023.*

C22 **Ziqiao Wang** and Yongyi Mao. "Information-Theoretic Analysis of Unsupervised Domain Adaptation." *International Conference on Learning Representations (ICLR), 2023.*

C23 **Ziqiao Wang** and Yongyi Mao. "On the Generalization of Models Trained with SGD: Information-Theoretic Bounds and Implications." *International Conference on Learning Representations (ICLR), 2022.*

Journal Articles

J1 Haiyun He, Yepeng Liu, Zhuoer Shen, **Ziqiao Wang**, Yongyi Mao, and Yuheng Bu. "Fundamental Limits of Statistical Watermarking for LLMs with Fixed and Optimizable Common Randomness." *IEEE Journal on Selected Areas in Information Theory (JSAIT). Major Revision.*

J2 Dezhi Liu, Richong Zhang, and **Ziqiao Wang**. "A word-context classification view of noise distributions in SGN-like representation learning." *Neurocomputing, 2027.*

Workshop Papers

W1 Wei Yao, Wenkai Yang, **Ziqiao Wang**, Yankai Lin, and Yong Liu. "Understanding the Capabilities and Limitations of Weak-to-Strong Generalization." *ICLR 2025 Workshop on Self-Improving Foundation Models Without Human Supervision (SSI-FM).*

W2 Haiyun He, Yepeng Liu, **Ziqiao Wang**, Yongyi Mao and Yuheng Bu. "Distributional Information Embedding: A Framework for Multi-bit Watermarking." *ICLR 2025 Workshop on GenAI Watermarking (WMark).*

W3 Haiyun He, Yepeng Liu, **Ziqiao Wang**, Yongyi Mao and Yuheng Bu. "Theoretically Grounded Framework for LLM Watermarking: A Distribution-Adaptive Approach." *ICLR 2025 Workshop on GenAI Watermarking (WMark).*

W4 **Ziqiao Wang** and Yongyi Mao. "On f-Divergence Principled Domain Adaptation: An Improved Framework." *18th Canadian Workshop on Information Theory (CWIT), 2024.*

W5 **Ziqiao Wang** and Yongyi Mao. "Two Facets of SDE Under an Information-Theoretic Lens: Generalization of SGD via Training Trajectories and via Terminal States." *Mathematics of Modern Machine Learning (M3L) Workshop at NeurIPS 2023.*

W6 Zixuan Liu*, **Ziqiao Wang***, Hongyu Guo and Yongyi Mao. "Over-Training with Mixup May Hurt Generalization." *First Workshop on Interpolation Regularizers and Beyond at NeurIPS 2022.*

Honors & Awards

2026 Top AC Award at NeurIPS 2026
Outstanding AC Award at ICLR 2026
2025 Shanghai Leading Young Talents (Overseas)
2024 Top 10% Reviewers Award at NeurIPS 2024
UAI 2024 Scholarship
Governor General's Academic Medal (Nominee for Spring 2025 Convocation)
Pierre Laberge Thesis Prize (Nominee for Spring 2025 Convocation)
2023 Top 10% Reviewers Award at NeurIPS 2023
ICLR 2023 Travel Award
2022-2024 INTER-MATH-AI Stipends from the NSERC CREATE program
2021-2024 Doctorate Admission Scholarship from University of Ottawa
2019-2024 International Doctoral Scholarship from University of Ottawa

Professional Service

Conference Reviewer NeurIPS (2023,2024,2025), ICML (2024,2025,2026), ICLR (2024,2025), AISTATS (2024,2025), UAI (2025), ACL (2026), AAAI (2025), IJCAI (2024), ISIT (2024, 2025), Neural Compression Workshop at ICML 2023

Journal Reviewer IEEE Transactions on Information Theory (TIT), Transactions on Machine Learning Research (TMLR), IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), IEEE Journal on Selected Areas in Information Theory (JSAIT), IEEE Transactions on Neural Networks and Learning Systems (TNNLS), Information and Inference: A Journal of the IMA (IMAIAI)

Area Chair ICLR, NeurIPS, ICML 2026 FoGen Workshop

Organizers The 2024 IEEE North American School of Information Theory (NASIT), The 2025 China Embodied AI Conference (CEAI), ITGenNexus Workshop at ISIT 2026

Teaching Experience

2025 – now **Instructor**, *Tongji University*.
CST6030 Principles of Artificial Intelligence (Spring 2026)
CST3526 Stochastic Process (Fall 2025)
CST2310 Introduction to Cyberspace Security (Fall 2025)

2018 – 2023 **Teaching Assistant**, *University of Ottawa*.
CSI 5138/5340[R00] Introduction to Deep Learning and Reinforcement Learning (2018-2019, 2021-2023)
ELG 5170[A00] Information Theory (2020)
GNG 1106[A00] Fundamentals of Engineering Computation (2020-2021, 2023)